Geometry Review for Electrodynamics

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Introduction

Electromagnetism can be understood, to a large extent, as a geometric theory.
Most of its structure follows directly from the geometry of spacetime, together with the existence of electric charge as a physical source.

This observation is important because the same geometric structures appear in many other field theories. Once the underlying geometry is understood, much of the formal structure of these theories becomes natural.

In this course we will therefore begin by reviewing the geometric framework in which classical field theories are formulated. The goal is to introduce the mathematical language needed to express the equations of electromagnetism in a coordinate–independent way.

The central idea is that physical laws should not depend on the particular coordinate system used to describe spacetime. Instead, they should be expressed in terms of geometric objects, such as vectors, covectors, and tensors, that are defined independently of coordinates.

Electromagnetism provides a particularly clear example of this principle. When formulated geometrically, Maxwell's equations take a very compact form and their structure becomes closely related to properties of spacetime itself.

For this reason we begin the course with a review of the geometry of spacetime and the basic algebraic structures needed to describe fields.

Why Geometry Appears in Field Theory

A classical field theory describes physical quantities that are defined at every point of spacetime. Examples include the electromagnetic field, temperature in a material, or the velocity field of a fluid.

Because fields are defined over spacetime, their mathematical description necessarily involves the geometric structure of spacetime itself. The equations that govern these fields must be formulated in a way that is independent of the choice of coordinates used to describe spacetime.

This requirement leads naturally to the use of geometric objects such as vectors, covectors, and tensors. These objects have well-defined transformation properties under changes of coordinates, which guarantees that physical laws retain the same form for all observers.

In practice, this means that the fundamental equations of a field theory should be written in terms of quantities that are defined geometrically rather than in terms of the components associated with a particular coordinate system.

Electromagnetism provides a clear illustration of this idea. When expressed in the usual three–vector notation, Maxwell's equations appear as four separate equations involving electric and magnetic fields. However, when written using tensorial objects defined on spacetime, these equations combine into a much simpler and more symmetric structure.

For this reason, before studying the electromagnetic field itself, we first review the geometric framework that allows us to express physical laws in a coordinate–independent way.

Metric and the Notion of Distance

In physics we constantly perform measurements. For example, we measure the distance between two points in space or the elapsed time between two events. To describe these measurements mathematically we need a rule that tells us how to assign lengths to displacements between nearby points.

This rule is provided by the **metric**.

Mathematically, a metric is a bilinear map that assigns a number to a pair of vectors. In particular, given a displacement vector v, the metric defines its squared length through

$$g(v, v).$$

In Euclidean space this rule reproduces the familiar Pythagorean expression. If a displacement has components $(\mathrm{d}x, \mathrm{d}y)$, the squared length is

$$\mathrm{d}s^2 = \mathrm{d}x^2 + \mathrm{d}y^2.$$

In spacetime the situation is different. Experiments in special relativity show that the combination of space and time intervals that remains invariant under changes of inertial reference frames is

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2.$$

This expression defines the **Minkowski metric**.

A crucial feature is that the time direction appears with a sign opposite to the spatial directions. The pattern of signs in the metric is called its **signature**. In this convention the signature is

$$(-, +, +, +).$$

This difference in sign is responsible for several distinctive features of relativistic spacetime, including the classification of intervals into timelike, spacelike, and null.

In coordinates, the Minkowski metric can be represented by the matrix

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Using this metric, the spacetime interval between two infinitesimally separated events can be written compactly as

$$\mathrm{d}s^2 = \eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu.$$

Unlike the Euclidean distance, this quantity can be positive, negative, or zero. These three possibilities correspond to different types of separations:

- **timelike:** $\mathrm{d}s^2 < 0$
- **spacelike:** $\mathrm{d}s^2 > 0$
- **null (lightlike):** $\mathrm{d}s^2 = 0$

The sign of the interval determines the causal relation between events. Timelike separations correspond to events that can influence each other through signals moving slower than light, while null separations correspond to propagation at the speed of light.

> **Note**
>
> The signature of a metric is invariant under coordinate transformations. Different coordinate systems may change the components of the metric, but the number of positive and negative directions remains the same.

This statement can be understood more concretely as follows. Any real symmetric bilinear form can be diagonalized by a suitable basis choice. After diagonalization, one can rescale basis vectors to reduce nonzero diagonal entries to $+1$ or -1. The only information that cannot be removed by such changes is the count of positive and negative signs. This is why the signature labels an invariant class of metrics rather than a particular matrix of components.

Vectors and the Metric

To understand the role of the metric more clearly, it is useful to think about displacement vectors.

Suppose we have two nearby points in space and consider the vector that connects them. This vector represents a displacement in space. In Euclidean geometry we can draw such a vector and measure its length using the metric.

The Euclidean metric allows us to compute not only the length of a vector, but also the relation between two vectors. Given two vectors v and u, the metric defines the quantity

$$g(v, u).$$

In Euclidean space this operation corresponds to the familiar **dot product**.

Geometrically, the dot product measures the projection of one vector along the direction of another. If u is a unit vector, the quantity

$$g(v, u)$$

gives the component of v along the direction u.

This interpretation is extremely useful. Instead of thinking of the metric only as a way of computing lengths, we can think of it as a rule that allows us to compare vectors and extract geometric information such as angles and projections.

In relativistic spacetime the same idea applies. The metric still provides a bilinear operation that takes two vectors and produces a number. The difference is that the sign structure of the metric is no longer Euclidean, reflecting the different roles of time and space in Minkowski geometry.

Points, Vectors, and Tangent Spaces

It is important to distinguish between points in spacetime and vectors.

We model spacetime as a manifold M equipped with a metric η. The elements of M are points (events), which represent locations in spacetime. In a coordinate system, a point $p \in M$ is described by numbers

$$(x^0, x^1, x^2, x^3).$$

These coordinates label the point, but they are not the point itself.

Vectors represent infinitesimal displacements between nearby points. A convenient and intrinsic way to define vectors is through their action on functions. Given the space of smooth functions on the manifold, $C^\infty(M)$ and $f(x^0, x^1, x^2, x^3) \in C^\infty(M)$, we define the action of a vector v at a point p as the directional derivative of f along v at p:

$$v(f) = \lim_{\epsilon \to 0} \frac{f(p + \epsilon v) - f(p)}{\epsilon}.$$

Here $p+\epsilon v$ is not literal addition of a point and a vector — that operation is not defined on a general manifold, as emphasized below. It is shorthand for any curve $\gamma : (-\delta, \delta) \to M$ through p with initial tangent v, i.e. a map from an interval of $\mathbb{R}$ into M satisfying $\gamma(0) = p$ and $\mathrm{d}x^\mu(\gamma(\epsilon))/\mathrm{d}\epsilon|_{\epsilon=0} = v^\mu$ in some (any) coordinate chart. Taylor-expanding such a curve in coordinates around $\epsilon = 0$ gives exactly

$$x^\mu(\gamma(\epsilon)) = x^\mu(p) + \epsilon\, v^\mu + O(\epsilon^2)$$

for small enough ϵ, which is what "$p + \epsilon v$" really denotes: the point reached after parameter distance ϵ along *some* curve realizing v at p, not a sum performed on M itself. Since the definition of $v(f)$ keeps only the $\epsilon \to 0$ limit, the result does not depend on which such curve is used; substituting the expansion above into $f(\gamma(\epsilon))$ and differentiating recovers the coordinate expression below directly, by the chain rule.

In coordinates, this becomes

$$v(f) = v^\mu \frac{\partial f}{\partial x^\mu}.$$

This definition captures the idea of a vector as a directional derivative operator and does not depend on a particular coordinate system.

Thus, we represent a vector as

$$v = v^\mu \partial_\mu, \qquad \text{with} \quad \partial_\mu \equiv \frac{\partial}{\partial x^\mu}.$$

The coordinate functions x^μ provide a natural basis $\{\partial_\mu\}$ for vectors at each point. It is important to note that the coordinates x^μ themselves are not vectors.

Spacetime is not, in general, a vector space: we cannot add points or multiply them by scalars. These operations are defined only for vectors.

At each point $p \in M$, we associate a vector space T_pM, called the tangent space at p. The elements of T_pM are the possible infinitesimal displacements starting from p.

Thus, points live in M, while vectors live in T_pM. Each point has its own tangent space, and there is no canonical identification between tangent spaces at different points.

In Minkowski spacetime, we can introduce global Cartesian coordinates. In this special case, it is possible to represent a displacement between two points p and q as

$$\Delta x^\mu = x^\mu(q) - x^\mu(p).$$

This makes it tempting to think of vectors as differences of points. In this setting, one can approximate the variation of a function as

$$f(q) - f(p) \approx \Delta x^\mu\, \partial_\mu f(p).$$

However, this construction relies on the existence of global Cartesian coordinates and a preferred origin. In a general curved spacetime, such coordinates do not exist, and the notion of a vector as a difference of points is not well defined.

For this reason, vectors must be defined intrinsically as elements of T_pM, independent of any coordinate system.

Spacetime can be viewed as a collection of points, each equipped with its own tangent space,

$$p \in M \quad \longrightarrow \quad T_pM.$$

Geometric objects such as vectors, covectors, and tensors are defined in these tangent spaces, not directly on M itself.

Covectors and the Dual Space

Given a vector space $\mathbb{V}$, we can construct another vector space called the dual space, denoted $\mathbb{V}^*$.

Elements of $\mathbb{V}^*$ are linear functionals, that is, maps

$$\tilde{w} : \mathbb{V} \to \mathbb{R}$$

such that for all $v_1, v_2 \in \mathbb{V}$ and scalars a, b,

$$\tilde{w}(av_1 + bv_2) = a\,\tilde{w}(v_1) + b\,\tilde{w}(v_2).$$

If $v \in \mathbb{V}$, the action of a covector $\tilde{w}$ on v produces a real number,

$$\tilde{w}(v).$$

This pairing between covectors and vectors is bilinear and is often denoted by

$$\langle \tilde{w}, v \rangle = \tilde{w}(v).$$

The set of all linear functionals forms a vector space, the dual space $\mathbb{V}^*$, and it has the same dimension as $\mathbb{V}$,

$$\dim \mathbb{V}^* = \dim \mathbb{V}.$$

However, $\mathbb{V}$ and $\mathbb{V}^*$ are distinct spaces. Without additional structure there is no canonical way to identify a vector with a covector.

This distinction becomes concrete once a basis is chosen. Let $\{e_i\}$ be a basis of $\mathbb{V}$. Then there exists a unique dual basis $\{\tilde{e}^i\}$ of $\mathbb{V}^*$ defined by

$$\tilde{e}^i(e_j) = \delta^i_j.$$

Any vector and covector can then be written as

$$v = v^i e_i, \qquad \tilde{w} = w_i \tilde{e}^i,$$

and their pairing becomes

$$\tilde{w}(v) = w_i v^i.$$

A map between $\mathbb{V}$ and $\mathbb{V}^*$ arises once a metric g is introduced. The metric is a bilinear map

$$g : \mathbb{V} \times \mathbb{V} \to \mathbb{R},$$

which allows us to associate to each vector v a covector $v^\flat \in \mathbb{V}^*$ via

$$v^\flat(w) = g(v, w).$$

In components, this corresponds to lowering an index,

$$v_i = \eta_{ij} v^j,$$

and the inverse metric provides the reverse operation (raising indices).

The dual space $\mathbb{V}^*$ itself has a dual, denoted $\mathbb{V}^{**}$. There is a natural map

$$\mathbb{V} \to \mathbb{V}^{**}, \qquad v \mapsto (\tilde{w} \mapsto \tilde{w}(v)),$$

which embeds vectors as linear functionals acting on covectors. In finite-dimensional spaces this map is an isomorphism, allowing us to identify $\mathbb{V}$ with $\mathbb{V}^{**}$.

In spacetime, at each point $p \in M$, we define the dual space T^*_pM as the space of covectors at p, that is, linear maps

$$T_pM \to \mathbb{R}.$$

A convenient way to construct covectors is through the exterior derivative. For a smooth function f, we define $\mathrm{d}f$ by its action on a vector v,

$$\mathrm{d}f(v) = v(f).$$

In coordinates, this gives

$$\mathrm{d}x^\mu(v) = v^\mu,$$

so the set $\{\mathrm{d}x^\mu\}$ forms a basis of T_p^*M dual to the basis $\{\partial_\mu\}$ of T_pM.

Tensor Products

Once vectors and covectors have been introduced, we can construct more general objects by combining them. The fundamental operation is the tensor product.

If $\mathrm{d}x^\mu$ and $\mathrm{d}x^\nu$ are covectors, their tensor product defines a bilinear map acting on two vectors:

$$(\mathrm{d}x^\mu \otimes \mathrm{d}x^\nu)(v, u) = \mathrm{d}x^\mu(v)\,\mathrm{d}x^\nu(u).$$

That is, each covector acts on one of the input vectors, and the results are multiplied, producing a real number.

More generally, given vector spaces $\mathbb{V}$ and $\mathbb{W}$, their tensor product $\mathbb{V} \otimes \mathbb{W}$ is defined so that bilinear maps on $\mathbb{V} \times \mathbb{W}$ correspond to linear maps on $\mathbb{V} \otimes \mathbb{W}$. In this way, tensor products allow us to encode multilinear operations within a linear framework.

Elements of $\mathbb{V}^* \otimes \mathbb{V}^*$ are called rank-two covariant tensors. In general, tensors are not single tensor products but linear combinations of them. For example,

$$T = \alpha\,\mathrm{d}x^0 \otimes \mathrm{d}x^1 + \beta\,\mathrm{d}x^3 \otimes \mathrm{d}x^3.$$

The objects $\mathrm{d}x^\mu \otimes \mathrm{d}x^\nu$ form a basis for this space, so any rank-two covariant tensor can be written as

$$T = T_{\mu\nu}\,\mathrm{d}x^\mu \otimes \mathrm{d}x^\nu.$$

This construction extends naturally to tensors of arbitrary type. By combining copies of $\mathbb{V}$ and $\mathbb{V}^*$, we obtain tensors of type (r, s):

$$T \in \underbrace{\mathbb{V} \otimes \cdots \otimes \mathbb{V}}_{r \text{ times}} \otimes \underbrace{\mathbb{V}^* \otimes \cdots \otimes \mathbb{V}^*}_{s \text{ times}}.$$

Such a tensor takes r covectors and s vectors as input and produces a real number in a multilinear way.

In spacetime, this construction is applied pointwise. At each point $p \in M$, we form tensors from the spaces T_pM and T_p^*M. A tensor field assigns to each point p a tensor in these spaces, varying smoothly across the manifold.

A central example is the metric, a rank-two covariant tensor field. In a coordinate basis, it can be written as

$$\eta = \eta_{\mu\nu}\,\mathrm{d}x^\mu \otimes \mathrm{d}x^\nu.$$

As a bilinear map, it acts on two vectors v and u as

$$\eta(v, u) = \eta_{\mu\nu} v^\mu u^\nu.$$

It can also be viewed as a map from vectors to covectors,

$$\eta(v) = \eta_{\mu\nu} v^\mu \mathrm{d}x^\nu,$$

which corresponds to lowering an index.

Since the metric is non-degenerate, it admits an inverse η^{-1} that raises indices and maps covectors to vectors. In coordinates,

$$\eta^{-1} = \eta^{\mu\nu}\,\partial_\mu \otimes \partial_\nu,$$

where $\eta^{\mu\nu}$ is the inverse matrix of $\eta_{\mu\nu}$. Its action on a covector is

$$\eta^{-1}(\mathrm{d}x^\mu) = \eta^{\mu\nu}\partial_\nu.$$

Symmetric and Antisymmetric Tensors

When tensor products involve multiple copies of the same vector space, it is useful to decompose tensors into symmetric and antisymmetric parts.

Consider a rank-two covariant tensor $T_{\mu\nu}$. Its symmetric and antisymmetric components are defined as

$$T_{(\mu\nu)} = \frac{1}{2}\left(T_{\mu\nu} + T_{\nu\mu}\right), \tag{1}$$

$$T_{[\mu\nu]} = \frac{1}{2}\left(T_{\mu\nu} - T_{\nu\mu}\right). \tag{2}$$

Any tensor can be uniquely written as the sum of these two parts,

$$T_{\mu\nu} = T_{(\mu\nu)} + T_{[\mu\nu]}.$$

The antisymmetric part is particularly important because it underlies the definition of the exterior (wedge) product.

Given two covectors w and k, their wedge product is defined by

$$w \wedge k = w \otimes k - k \otimes w,$$

which is manifestly antisymmetric. Evaluated on basis vectors, it gives

$$(w \wedge k)_{\mu\nu} \equiv (w \wedge k)(\partial_\mu, \partial_\nu) = w_\mu k_\nu - w_\nu k_\mu = 2\, w_{[\mu} k_{\nu]}.$$

On the other hand, when the wedge product is expanded in the basis of 2-forms, we write

$$w \wedge k = w_{[\mu} k_{\nu]}\, \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu,$$

where the antisymmetrization brackets include the factor $1/2$. The apparent difference by a factor of 2 reflects the distinction between the components of a tensor in a basis and its evaluation as a bilinear map.

More generally, the wedge product corresponds to the antisymmetrization of a tensor product and produces tensors that vanish whenever two arguments are equal.

The antisymmetric product has a direct geometric interpretation. Given two vectors v and u, the object $v \wedge u$ represents an oriented area element.

If the vectors are linearly dependent, the parallelogram they span has zero area, and correspondingly

$$v \wedge u = 0.$$

If they are linearly independent, $v \wedge u$ is nonzero and encodes both the magnitude and orientation of the area they define.

This interpretation extends to higher degrees: the wedge product of three vectors represents an oriented volume, and in general, antisymmetric tensors of degree p represent oriented p-dimensional volume elements. In an n-dimensional space, the highest nonvanishing antisymmetric object involves n vectors.

These antisymmetric structures are central in physics, as they provide a natural and coordinate-independent way to describe areas, volumes, and fluxes.

Antisymmetric Products and Linear Independence

Antisymmetric products provide a powerful way to describe geometric structures without introducing additional concepts such as a metric.

To illustrate this, consider two vectors w and k. Suppose we want to determine whether they are linearly independent. Normally this requires solving an equation of the form

$$\alpha w + \beta k = 0.$$

If the only solution is $\alpha = \beta = 0$, the vectors are linearly independent.

The antisymmetric product offers an alternative way to test this. If the wedge product of the two vectors is nonzero,

$$w \wedge k \neq 0,$$

then the vectors must be linearly independent.

Geometrically this statement is very natural. Two independent vectors span an area, and the wedge product represents the **oriented area element** generated by them. If the vectors are parallel, the area they span is zero, and the wedge product vanishes.

This idea extends naturally to higher dimensions. For example, given four vectors in a four-dimensional space we can form

$$w_1 \wedge w_2 \wedge w_3 \wedge w_4.$$

If this object is nonzero, the four vectors span a four-dimensional volume and are therefore linearly independent.

More generally, in an n-dimensional space the largest nonzero antisymmetric product involves n vectors. Any antisymmetric product containing more than n vectors must vanish, because at least two of the vectors must be linearly dependent.

This property reflects the fact that an n-dimensional space cannot contain geometric objects with dimension larger than n.

Differential Forms and Integration

The antisymmetric tensors we introduced have an important application in the definition of integration on geometric objects such as curves and surfaces.

Consider a surface in space that is parametrized by two parameters, λ_1 and λ_2. Each point of the surface can then be written as a function of these parameters,

$$x^\mu = x^\mu(\lambda_1, \lambda_2).$$

If we vary one parameter while keeping the other fixed, we obtain a vector tangent to the surface. For example,

$$\frac{\partial x^\mu}{\partial \lambda_1}, \qquad \frac{\partial x^\mu}{\partial \lambda_2}$$

are two tangent vectors that span the surface locally.

These two vectors define an infinitesimal oriented area element. To measure the size of this area element we use an antisymmetric tensor acting on the pair of tangent vectors.

Objects of this type are called differential forms. A two-form takes two vectors as input and produces a number, and its antisymmetry ensures that it measures an oriented area element.

Integration can then be understood geometrically as summing the contributions of such infinitesimal elements. For example, if ω is a two-form and Σ is a surface, the integral

$$\int_\Sigma \omega$$

is obtained by evaluating the form on the tangent vectors generated by the parametrization and summing the result over the surface.

This viewpoint makes the definition of integration independent of the choice of coordinates. The parametrization simply provides the tangent vectors, while the differential form encodes the geometric quantity being measured.

An equivalent way to say this is that integration uses the pullback of the form to the parameter domain. Changing parametrization modifies Jacobian factors and tangent vectors in compensating ways, so the geometric integral over the same oriented domain is unchanged.

The Exterior Derivative

We now introduce an operator that allows us to differentiate geometric objects in a coordinate-independent way. We introduced it above as a way to construct covectors from scalar functions, but it can be extended to act on differential forms of any degree. This operator is called the exterior derivative and is denoted by d.

The simplest case is the action of d on a scalar function f. The result is a covector, written

$$\mathrm{d}f.$$

This covector acts on a vector v by producing the directional derivative of f along that vector,

$$\mathrm{d}f(v) = v(f).$$

In other words, $\mathrm{d}f$ measures how the function changes when we move in the direction specified by v.

In coordinates, if the vector v has components v^μ, the action of the covector $\mathrm{d}f$ can be written as

$$\mathrm{d}f(v) = v^\mu \partial_\mu f.$$

This expression shows explicitly that $\mathrm{d}f$ contains the derivatives of the function with respect to the coordinates. In this sense, $\mathrm{d}f$ corresponds to the familiar gradient of a scalar function, but interpreted as a covector.

The exterior derivative extends naturally to differential forms of higher degree, producing new forms whose antisymmetric structure encodes derivatives in a coordinate-independent way. This construction will allow us to write the equations of electromagnetism in a compact geometric form.

The Exterior Derivative and Stokes' Theorem

The exterior derivative allows us to connect differentiation with integration in a natural geometric way.

If f is a scalar function, its exterior derivative $\mathrm{d}f$ is a covector. As we have seen, this covector acts on a vector v by producing the directional derivative of f,

$$\mathrm{d}f(v) = v(f).$$

Now consider integrating this object along a curve. If the curve is parametrized by a coordinate x, we can write

$$\int \mathrm{d}f.$$

Using the definition of the exterior derivative, this integral becomes

$$\int \frac{\partial f}{\partial x}\mathrm{d}x.$$

This is simply the ordinary integral of a derivative. Therefore,

$$\int_a^b \mathrm{d}f = f(b) - f(a).$$

This result is familiar from basic calculus, but it has a deeper geometric meaning. It shows that the integral of a derivative over a region depends only on the values of the function on the **boundary** of that region.

This idea generalizes to higher dimensions. When applied to differential forms, the exterior derivative leads to the general statement known as **Stokes' theorem**:

$$\int_{\partial\Omega} \omega = \int_{\Omega} \mathrm{d}\omega.$$

Here Ω is a region of space and $\partial\Omega$ is its boundary. The form ω is integrated over the boundary, while its exterior derivative $\mathrm{d}\omega$ is integrated over the interior.

This theorem provides a unified geometric framework for many familiar results from vector calculus, including the divergence theorem and the classical version of Stokes' theorem.

In the context of electromagnetism, these relations play a central role because Maxwell's equations can be written compactly in terms of differential forms and their exterior derivatives.

Gradients as Covectors

Given a scalar function $f(x)$, its exterior derivative $\mathrm{d}f$ defines a covector field with components

$$(\mathrm{d}f)_\mu = \partial_\mu f.$$

This object acts on a vector $v = v^\mu \partial_\mu$ by

$$\mathrm{d}f(v) = v^\mu \partial_\mu f,$$

which is the directional derivative of f along v. In this sense, $\mathrm{d}f$ encodes how f changes in different directions and is naturally a covector.

In Euclidean space, or more generally in the presence of a metric η, we can associate to this covector a vector by raising the index. This defines the gradient of f as the vector field

$$\eta^{-1}(\mathrm{d}f) = \eta^{\mu\nu}\partial_\nu f\, \partial_\mu.$$

Thus, the familiar gradient is not a fundamentally new object, but rather the vector obtained from the covector $\mathrm{d}f$ using the metric. Without a metric, $\mathrm{d}f$ is the natural geometric object, while the notion of a gradient as a vector depends on the additional structure provided by η.

Coordinate Dependence of the Metric

The components of the metric tensor depend on the coordinate system used to describe spacetime. However, the scalar quantity

$$g(v, u)$$

is invariant.

Different coordinate systems therefore lead to different matrices representing the metric, even though the underlying geometric structure remains the same.

This idea leads to the principle of **covariance**, according to which physical laws should be written in a form that remains valid under arbitrary changes of coordinates.

In flat spacetime, invariance under Lorentz transformations is an important special case of this broader principle. The covariant formulation makes clear which statements are geometric and therefore independent of any chosen frame.

Covariant Observer Decomposition and Rapidity

For relativistic field theory, it is useful to decompose tensors relative to an observer with unit timelike four-velocity u^μ satisfying

$$\eta_{\mu\nu}u^\mu u^\nu = -1.$$

The projector onto the local spatial subspace orthogonal to u^μ is

$$h_{\mu\nu} = \eta_{\mu\nu} + u_\mu u_\nu,$$

which satisfies

$$h_{\mu\nu}u^\nu = 0, \qquad h^\mu{}_\alpha h^\alpha{}_\nu = h^\mu{}_\nu.$$

Thus $h^\mu{}_\nu$ removes the temporal part (relative to u^μ) and keeps the spatial part.

Given another timelike unit vector w^μ, we can decompose it as

$$w^\mu = -(u\cdot w)\,u^\mu + h^\mu{}_\nu w^\nu.$$

The scalar $-(u\cdot w)$ and the spatial norm $\sqrt{h_{\mu\nu}w^\mu w^\nu}$ are naturally parametrized by a rapidity ξ:

$$-(u\cdot w) = \cosh\xi, \qquad \sqrt{h_{\mu\nu}w^\mu w^\nu} = |\sinh\xi|\,.$$

In frames where relative speed v is defined, this gives

$$\tanh\xi = \frac{v}{c}, \qquad \cosh\xi = \gamma, \qquad \gamma = \frac{1}{\sqrt{1-v^2/c^2}}.$$

This decomposition is central in covariant electromagnetism: electric and magnetic fields are observer-dependent spatial projections of a single spacetime tensor.

In this language, the magnetic part is fundamentally area-like (bivector-like), not intrinsically a standard vector. The familiar pseudovector description of $\mathbf{B}$ is a special feature of three spatial dimensions, where oriented areas can be identified with directions.

Exercises

Exercise 1: Signature and the Minkowski Metric (Essential)

a) For the Lorentz transformation $t' = \gamma(t - vx/c^2)$, $x' = \gamma(x - vt)$, $y' = y$, $z' = z$, show explicitly that $-c^2\mathrm{d}t'^2 + \mathrm{d}x'^2 + \mathrm{d}y'^2 + \mathrm{d}z'^2$ equals $-c^2\mathrm{d}t^2 + \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2$.

b) Show that any timelike vector v^μ can always be rescaled so that $\eta_{\mu\nu}v^\mu v^\nu = -1$, and explain the physical meaning of this normalization (it is exactly the condition satisfied by a four-velocity).

Exercise 2: Orthogonality and Non-Degeneracy

a) Two vectors are orthogonal if $g(u,v) = 0$. Give an explicit example of two nonzero vectors in Minkowski space that are orthogonal in this sense but are *not* perpendicular in the ordinary Euclidean sense (consider a null vector).

b) Show that if a bilinear form g satisfies $g(v,u) = 0$ for every u, then $v = 0$. Explain why this property is called non-degeneracy, and why it is what allows the metric to be inverted.

Exercise 3: Tangent Vectors and Coordinate Changes (Essential)

a) For a curve $x^\mu(\lambda)$, show that under a reparametrization $\lambda \to \lambda'(\lambda)$ the tangent vector $v^\mu = \mathrm{d}x^\mu/\mathrm{d}\lambda$ transforms as $v'^\mu = (\mathrm{d}\lambda/\mathrm{d}\lambda')\,v^\mu$.

b) Using the directional-derivative definition $v(f) = v^\mu\partial_\mu f$, show that the coordinate basis vectors $\{\partial_\mu\}$ are linearly independent and span T_pM, i.e. that they form a basis.

c) Under a coordinate change $x^\mu \to x'^\mu(x)$, show that the basis vectors transform as $\partial_\mu = (\partial x'^\nu/\partial x^\mu)\,\partial'_\nu$.

Exercise 4: The Dual Basis and the Exterior Derivative

a) Given a basis $\{e_i\}$ of $\mathbb{V}$, construct the dual basis $\{\tilde{e}^i\}$ explicitly and show that it is unique.

b) Show that $\mathrm{d}x^\mu(\partial_\nu) = \delta^\mu_\nu$, and explain why this is exactly what is needed for $\{\mathrm{d}x^\mu\}$ to be the basis of T^*_pM dual to $\{\partial_\mu\}$.

c) Using only $\mathrm{d}f(v) = v(f)$, show that $\mathrm{d}(fg) = f\,\mathrm{d}g + g\,\mathrm{d}f$.

Exercise 5: The Metric as a Map Between Vectors and Covectors (Essential)

a) For $v = v^\mu\partial_\mu$, compute the components of the covector $\eta(v)$.

b) Show that applying η^{-1} to $\eta(v)$ recovers the original vector v — i.e. that lowering an index and then raising it is the identity.

Exercise 6: Wedge Products, Area, and Linear Independence (Essential)

a) For two vectors v, u, show that $v \wedge u = 0$ if and only if v and u are linearly dependent.

b) Show that in an n-dimensional space, any wedge product of more than n vectors must vanish.

c) Using the interpretation of $v \wedge u$ as an oriented area element, explain in your own words why (a) and (b) are geometrically obvious rather than merely algebraic coincidences.

Exercise 7: The Exterior Derivative and Stokes' Theorem (Essential)

a) Compute $\mathrm{d}f$ explicitly for $f(x, y, z) = x^2yz$.

b) Show that $\mathrm{d}^2f = 0$ for any smooth scalar function f (this relies on partial derivatives commuting).

c) Show that the fundamental theorem of calculus, $\int_a^b \mathrm{d}f = f(b) - f(a)$, is precisely the one-dimensional case of Stokes' theorem $\int_{\partial\Omega}\omega = \int_\Omega \mathrm{d}\omega$.

Exercise 8: Gradients and the Role of Coordinates

a) In Euclidean space, verify that raising the index of $\mathrm{d}f$ with δ^{ij} reproduces the usual gradient vector ∇f.

b) Starting from $\mathrm{d}s^2 = \mathrm{d}x^2 + \mathrm{d}y^2$, compute the metric components in polar coordinates $x = r\cos\theta$, $y = r\sin\theta$, and verify that the scalar product $g(v, u)$ of two fixed vectors is unchanged by the coordinate change.

c) Explain why the matrix representation of the metric changes under this coordinate transformation even though the underlying geometry does not.

Exercise 9: Observer Decomposition, Rapidity, and the Electric/Magnetic Split (Essential)

a) Show explicitly that $h_{\mu\nu}u^\nu = 0$ and that $h^\mu{}_\nu$ is a projector, i.e. $h^\mu{}_\alpha h^\alpha{}_\nu = h^\mu{}_\nu$.

b) For a vector v^μ, define $v^\mu_\parallel = -(u\cdot v)\,u^\mu$ and $v^\mu_\perp = h^\mu{}_\nu v^\nu$. Show that $v^\mu = v^\mu_\parallel + v^\mu_\perp$ and that $u_\mu v^\mu_\perp = 0$.

c) For two unit timelike vectors u^μ, w^μ, show that $-(u\cdot w) \geq 1$, and using $-(u\cdot w) = \cosh\xi = \gamma$, interpret this bound physically.

d) For a rank-two antisymmetric tensor $F_{\mu\nu}$, define $E_\mu \equiv F_{\mu\nu}u^\nu$ and $B^\mu \equiv \frac{1}{2}\epsilon^{\mu\nu\alpha\beta}u_\nu F_{\alpha\beta}$. Show that both are orthogonal to u^μ, and explain why changing the observer u^μ changes this electric/magnetic split even though $F_{\mu\nu}$ itself does not.